Employee Salary Prediction

Machine Learning Project — Full Documentation

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1. Problem Statement

Background

Organizations spend significant time and resources manually assessing compensation structures for their workforce. Salary decisions are often based on subjective judgment, inconsistent criteria across departments, or outdated benchmarks. This leads to inequity, budget overruns, and employee dissatisfaction.

The Challenge

HR departments and management at a mid-sized manufacturing and services company operating across Egypt, Saudi Arabia, UAE, Syria, and Lebanon need a data-driven method to:

- Predict employee annual salary based on measurable factors
- Standardize compensation decisions across departments and locations
- Identify what drives salary variation in their workforce

Objective

Build a machine learning model that predicts an employee's annual salary using years of experience and job performance rating, deploy it as an interactive web application, and generate actionable insights from the data through exploratory data analysis.

Success Criteria

- A trained regression model with measurable accuracy (MAE metric)
- A deployed Streamlit web app allowing real-time salary predictions
- Clear visualizations revealing salary trends across departments, countries, and centers

2. Dataset Overview

The dataset (Employees.xlsx) contains 689 employee records with 15 features, representing staff across 5 countries and multiple departments.

Feature	Type	Description
No	Integer	Employee ID
First Name / Last Name	String	Employee name
Gender	Categorical	Male / Female
Start Date	Date	Date of joining
Years	Integer	Years at the company
Department	Categorical	Business unit / team
Country	Categorical	Egypt, Saudi Arabia, UAE, Syria, Lebanon
Center	Categorical	Branch: Main, North, South, West
Monthly Salary	Integer	Monthly pay (USD)

Feature	Type	Description
Annual Salary	Integer	🎯 Target variable — yearly pay
Job Rate	Float (1–5)	Performance rating
Sick Leaves	Integer	Days of sick leave
Unpaid Leaves	Integer	Days of unpaid leave
Overtime Hours	Integer	Total overtime hours worked

Data Quality: ✅ No missing values (0 nulls across all 15 columns) | ✅ No duplicate rows

3. Exploratory Data Analysis (EDA)

3.1 Gender Distribution

• Male: 449 employees (65.2%)

• Female: 240 employees (34.8%)

The workforce is predominantly male. This is worth monitoring from an HR equity standpoint.

3.2 Job Rate Distribution

• Mean: 3.59 | Std: 1.35 | Min: 1.0 | Max: 5.0

• Most employees cluster at ratings 3 and 5, suggesting a bimodal distribution.

3.3 Overtime Hours

• Mean: 13.7 hours | Median: 7 hours | Max: 198 hours

• Heavily right-skewed — a small number of employees work extreme overtime (up to 198 hrs). This may indicate outliers or specific high-demand roles.

3.4 Annual Salary Distribution

• Mean: $24,818 | Min: $8,436 | Max: $41,400

• Salary range spans ~$33K, indicating significant variation across roles, seniority, and departments.

3.5 Salary by Department

Top-paying departments (by average annual salary) were identified via bar chart. This helps detect compensation inequity across business units.

3.6 Salary by Center & Job Rate by Country

Geographic differences in both salary and performance ratings are visible across the 4 centers (Main, North, South, West) and 5 countries.

4. Machine Learning Model

4.1 Feature Selection

Two features were selected as model inputs:

• Years — seniority/experience at the company

• Job Rate — employee performance rating (1–5)

Target variable: Annual Salary (continuous numeric value)

4.2 Train / Test Split

• Training set: 551 samples (80%)

• Test set: 138 samples (20%)

4.3 Algorithm: Linear Regression

Linear Regression is the foundational supervised learning algorithm for predicting continuous values. It models the relationship between features and the target as a linear equation:

$$\text{Annual Salary} = \beta_0 + \beta_1(\text{Years}) + \beta_2(\text{Job Rate})$$

It was chosen as a starting model because it is interpretable, fast to train, and provides a strong baseline.

4.4 Model Performance

Metric	Value
Mean Absolute Error (MAE)	~$7,703
Salary Range in Data	$8,436 – $41,400
Error as % of Range	~23.3%

The MAE of $7,703 represents a reasonable baseline for a 2-feature model. Given the wide salary range ($33K spread), adding more features is expected to reduce error significantly.

4.5 Model Saving

The trained model was serialized using joblib and saved as linearmodel.pkl for use in the Streamlit web application.

5. Web Application (Streamlit)

The app.py file implements a fully interactive Streamlit web application that loads the saved model and allows HR teams or managers to predict any employee's salary in real time.

App Features:

• Number input for Years of experience (minimum: 0)

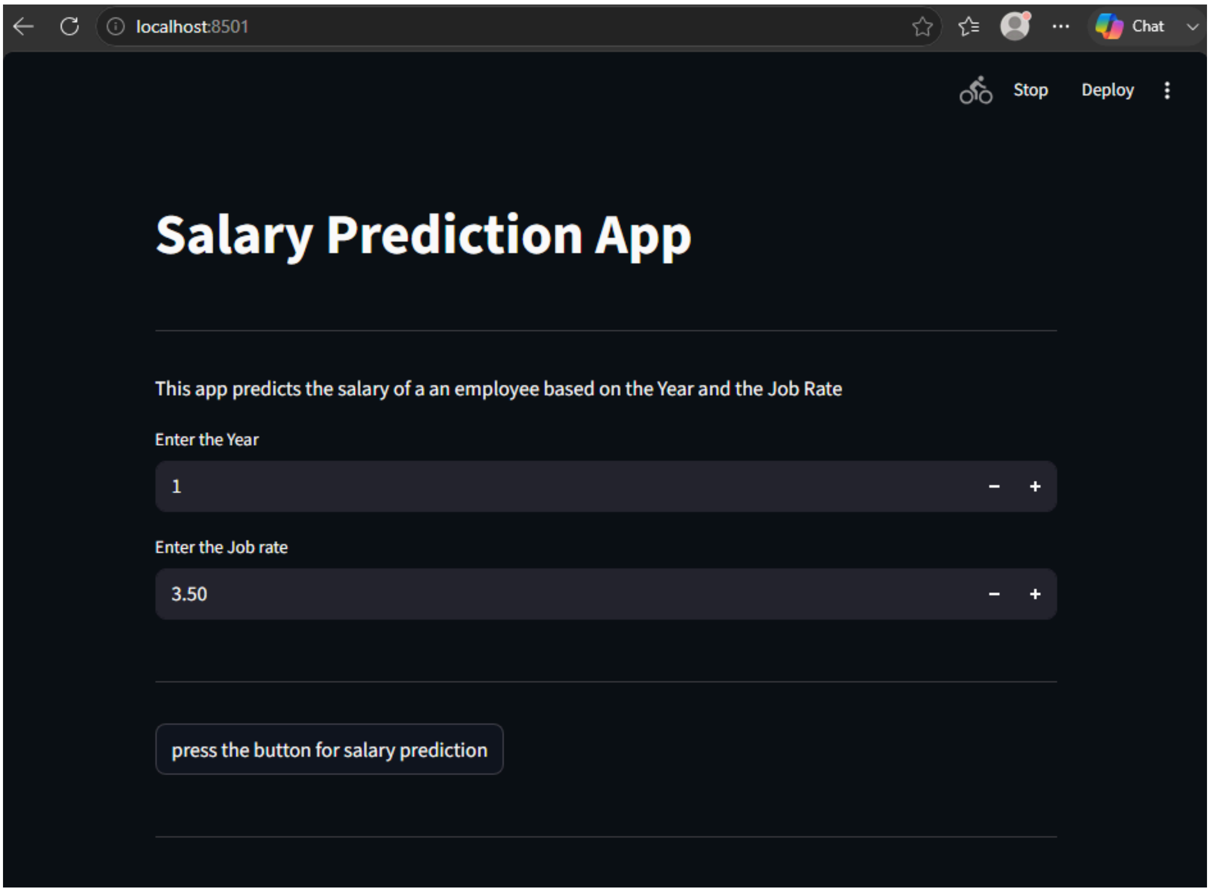
• Number input for Job Rate (0.0 – 5.0, step 0.5)

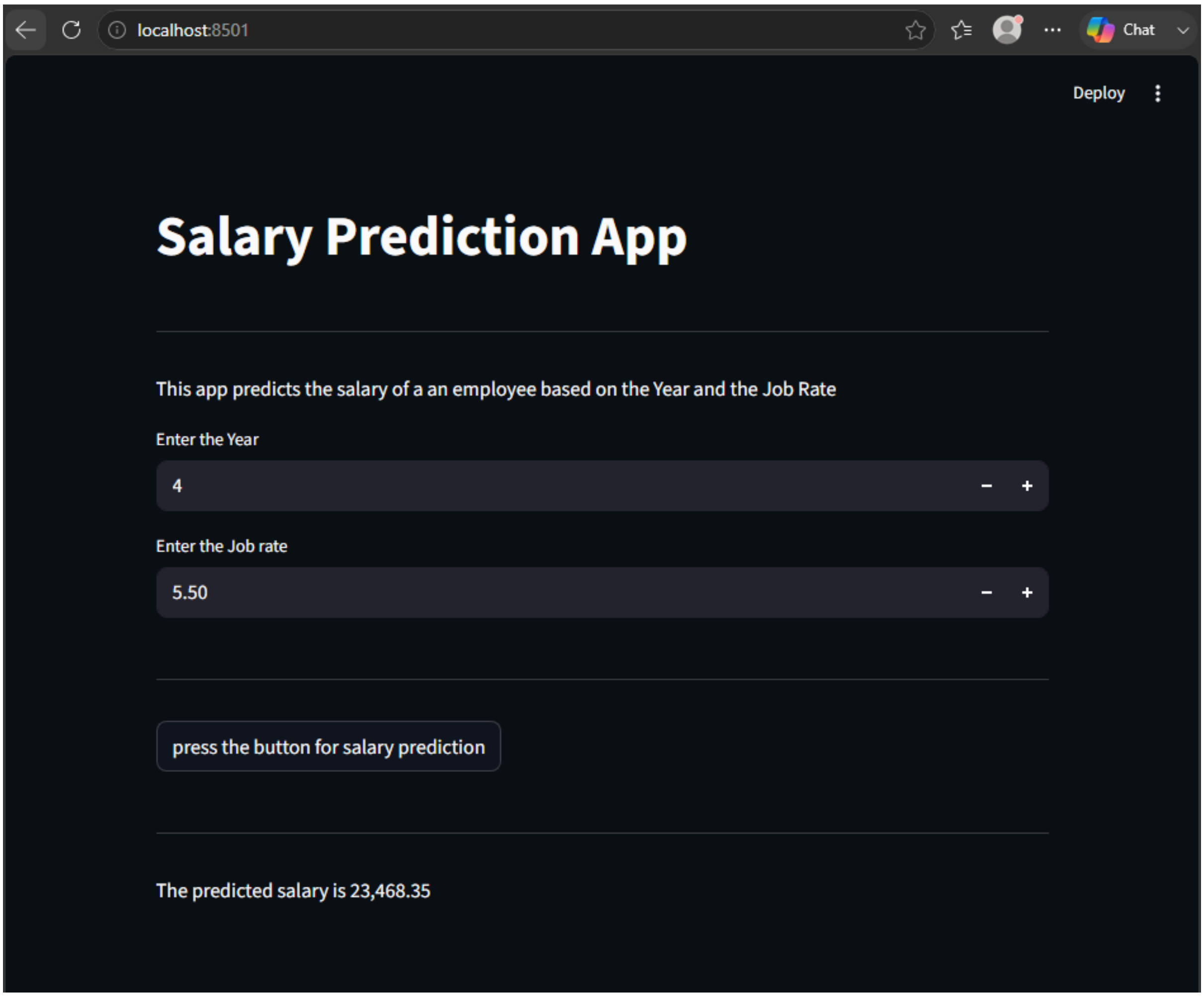
• Predict button triggers model inference

• Displays formatted predicted salary with balloon animation

To run the app:

```
streamlit run app.py
```


localhost:8501
Chat
Stop
Deploy
Salary Prediction App
This app predicts the salary of a an employee based on the Year and the Job Rate
Enter the Year
1
Enter the Job rate
3.50
press the button for salary prediction



6. Limitations & Future Improvements

Current Limitations

- Only 2 features used — significant salary variation remains unexplained
- Linear Regression assumes a linear relationship which may not hold fully
- No cross-validation used — single train/test split can give unstable estimates

Recommended Improvements

- Add features: Department, Country, Overtime Hours, Center
- Encode categorical variables with One-Hot Encoding
- Try Random Forest or XGBoost for non-linear modeling
- Add k-fold cross-validation for more reliable MAE
- Deploy app to Streamlit Cloud or Hugging Face Spaces

7. Tech Stack

• Language: Python 3
• Data: pandas, numpy, openpyxl
• Visualization: matplotlib, seaborn
• Machine Learning: scikit-learn (LinearRegression, train_test_split, MAE)
• Model Persistence: joblib
• Web App: Streamlit
• Notebook Environment: Jupyter Notebook

GitHub: https://github.com/AngeloGeno/Machine-learning-Salary-Prediction